

Discrete Exterior Calculus

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Exterior Calculus (EC) [1], also known as the calculus of differential forms, is a powerful mathematical framework for formulating and analyzing problems involving geometry and topology. Developed to generalize classical vector calculus, EC provides a coordinate-free approach particularly suited for manifold settings.

Central in EC is the concept of *differential form*. Basically, a differential k -form is an object that can be integrated over k -dimensional manifolds. The primary operations in EC include:

- The **exterior derivative** d : a generalization of gradient, curl, and divergence.
- The **wedge product** $\wedge$: an associative, bilinear, and antisymmetric product of forms.
- The **Hodge star** $\star$: maps k -forms to $(n - k)$ -forms on an n -dimensional oriented Riemannian manifold.

EC provides a general and unified approach, which is particularly useful in fields such as differential geometry, topology, and theoretical physics. Table 1 shows a comparison between vector and exterior calculus.

While EC provides a powerful and elegant continuous framework, a discrete counterpart is needed for practical applications (e.g., in computational physics). Discrete Exterior Calculus (DEC) [2, 3] is exactly the discrete version of EC. It operates on simplicial complexes (which play the role of the manifolds for EC) and maintains the geometric and topological structure of the continuous counterpart. Most importantly, DEC is inherently a discrete framework that bypasses the need for a differential formulation of physical theories. As a result, local conservation laws are naturally and strongly enforced, without requiring the discretization of differential operators [4]. In [5] the authors show that DEC can be specialized into other well-known numerical methods, such as Finite Volume Method and Finite Differences. Table 2 shows a comparison with DEC and the Finite Element Method (FEM), another popular choice for the discretization of Partial Differential Equations (PDE).

Feature	Vector Calculus	Exterior Calculus
Differentiation	∇ , curl, div	Exterior derivative d
Integration domain	$\Omega \subset \mathbb{R}^n$	n -manifolds
Variables	Vector fields	Differential forms
Product	Dot/cross products	Wedge product $\wedge$
Stokes' theorem	Several variants	$\int_{\partial\Omega} \omega = \int_{\Omega} d\omega$
Operator identity	$\nabla \cdot (\nabla \times \mathbf{F}) = 0$	$d^2 = 0$
Hodge star	Not used explicitly	Maps between dual degrees
Duality	Vector/tensor duality	$\star$ duality of forms

Table 1: Overview of the key differences between vector and exterior calculus.

This chapter is organized as follows. Section 1 presents the core concepts of DEC. Section 2 collects some examples of application of DEC for different scenarios. Finally, Section ?? gives a gentle and concise introduction to the discrete calculus library `dctkit`, which provides a python implementation of the concepts described in this chapter.

Let us fix some notations used in the following sections:

- $\mathcal{E}^n$ is an n -dimensional euclidean space.
- $\mathcal{V}$ is the n -dimensional translation space of $\mathcal{E}^n$ with base $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$.
- $\langle\langle \cdot, \cdot \rangle\rangle : \mathcal{V} \times \mathcal{V} \rightarrow \mathbb{R}$ is an inner product, i.e. a symmetric bilinear positive definite function.

Feature	FEM	DEC
Mathematical foundation	Variational principles	Exterior calculus
Variables	Scalar/vector fields in H^1	Cochains
Mesh structure	Triangular/quadrilateral	Simplicial complex
Differentiation	Weak derivatives	Coboundary
Metric dependency	In shape functions and integrals	In the Hodge star
Topological invariants	Not explicitly preserved	Naturally preserved
Local conservation laws	Enforced in weak form	Strongly enforced

Table 2: Overview of the key differences between the Finite Element Method (FEM) and Discrete Exterior Calculus (DEC).

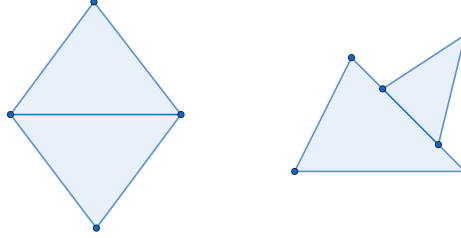


Figure 1: An example of a 2-simplicial complex (left) and a not simplicial complex (right).

1 DEC FUNDAMENTALS

In this section we provide a gentle introduction about the fundamentals of Discrete Exterior Calculus (DEC). For further details on DEC, the reader is referred to [2, 6, 3, 7, 8].

1.1 Discrete Domains

Definition 1. A p -cell σ^p is defined as a set of points which is homeomorphic to a closed unit p -ball $B_p := \{x \in \mathbb{R}^p \mid \|x\| \leq 1\}$. The boundary of the p -cell $\partial\sigma^p$ is the portion of the cell which is mapped through the homeomorphism to ∂B_p . When a p -cell consists of exactly $p + 1$ vertices it is called p -simplex.

We can think of a cell complex as the discrete counterpart of a manifold, in the sense that, within discrete calculus, it plays a role analogous to that of a manifold in classical exterior calculus. In fact, the following definition corresponds exactly to the concept of a discrete manifold in [6] (see p. 43, Section 3.1.5). In [2], such structures are referred to as *manifold-like cell complexes*.

Definition 2. A collection of cells defines a **cell complex** if it satisfies the following rules:

- The boundary of each p -cell (for $p > 0$) comprises the union of lower-order p -cells.
- The intersection of any two cells is either empty or a boundary element of both cells.

A **simplicial complex** is a cell complex in which every p -cell is a p -simplex. In Figure 1 we report an example of a 2-simplicial complex alongside an example of a non-simplicial complex. Notice that the latter is not a simplicial complex because the intersection of two 1-cells is not a boundary element of both cells.

We now introduce some operative definitions that we are going to use in the following sections.

Definition 3. A **well-centered simplicial complex** is a simplicial cell-complex where the circumcenter of any p -simplex of the complex lies in its interior.

We will require that each p -cell is oriented and the orientation is specified by the order of nodes used to represent the p -cell. See p. 42-44 of [8] for more details in this regard.

Definition 4. We also introduce the following notation for p -simplices: $\sigma^q \prec \sigma^p$ if and only if σ^q is a proper face of σ^p .

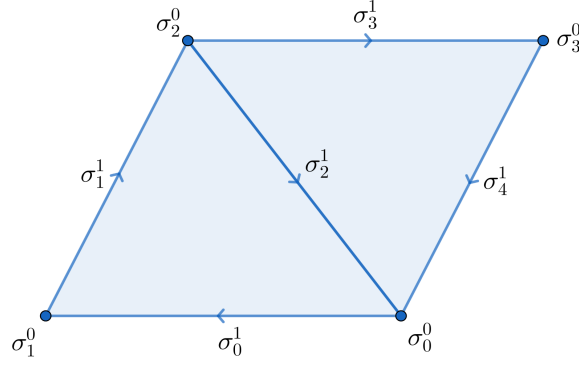


Figure 2: The 2-simplicial complex of Example 1.

Definition 5. The **incidence matrix** A_p is the matrix which encodes which p -cells are incident to which $(p-1)$ -cells in the n -complex and it is defined as

$$A_p(i, j) = \begin{cases} 0 & \text{if } \sigma_i^{p-1} \text{ is not on the boundary of } \sigma_j^p, \\ +1 & \text{if } \sigma_i^{p-1} \text{ is coherent with the induced orientation of } \sigma_j^p, \\ -1 & \text{if } \sigma_i^{p-1} \text{ is not coherent with the induced orientation of } \sigma_j^p. \end{cases} \quad (1)$$

Example 1. Let us see an explicit example to understand better the previous definition of incidence matrix. Consider the simplicial complex in Figure 2. In this case, the incidence matrix A_1 is

$$A_1 = \begin{bmatrix} -1 & 0 & 1 & 0 & 1 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

Definition 6. Let K be a cell complex. A p -**chain** is an n_p -tuple of scalars which assigns a coefficient to each p -cell, where n_p is the number of distinct p -cells in the complex. Each p -cell is referred to a **basic chain**. The p -chains group is denoted by $C_p(K, \mathbb{R})$.

A p -chain c_p can be expressed with respect to basic chains:

$$c_p = \sum_j c_p(\sigma_j^p) \sigma_j^p, \quad (2)$$

where $c_p(\sigma_j^p) \in \mathbb{R}$ and σ_j^p is a basic chain for any j . Thus, any p -chain may be represent by a $n_p \times 1$ (row) vector c_p . By applying the incidence matrix A_p to c_p we obtain a $(p-1)$ -chain, i.e., in components

$$c_{p-1} = A_p c_p, \quad (3)$$

where c_{p-1} is the array of the coefficients of the **boundary** of the p -chain c_p . Hence, A_p may be viewed as the matrix representation of the **discrete boundary operator**. We will indicate the boundary operator with the symbol ∂ . Hence, the incidence matrix provides both a representation of the topology of the manifold and a representation of the boundary operator.

Geometrically, p -chains are the discrete representation of p -vectors: their coefficients (weights) can be used to encode length, areas, etc. For instance, the 1-chains represent lengths, while the 2-chains represent areas and so on.

1.2 Discrete Forms and Coboundary

Definition 7. A p -**cochain**, or discrete form, is a linear functional that maps chains to scalars. As for the chains, we can represent cochains in terms of **basic cochains**

$$c^p = \sum_i c^p(\sigma_p^i) \sigma_p^i, \quad (4)$$

where $a_i \in \mathbb{R}$ and σ_p^i is a basic p -cochain. The space of p -cochains on a simplicial complex K is denoted by $C^p(K, \mathbb{R})$.

Cochains, or *discrete forms*, serve as the discrete analogue of differential forms, and in physical theories they represent discrete fields defined on the elements of a cell complex. Indeed, we know [8] that differential p -forms are built taking the dual of the vector space of p -vectors and similarly the space of scalar-valued p -cochains is the dual of the space of scalar-valued p -chains. We can represent a p -cochain as a $1 \times n_p$ (column) vector.

Definition 8. We can define a discrete analogue of **integration** as the pairing of a p -chain with a p -cochain, precisely

$$\langle\langle c^p, c_p \rangle\rangle := \sum_{i=1}^{n_p} c_p(\sigma_i^p) c^p(\sigma_p^i). \quad (5)$$

Definition 9. We define the matrix form of **coboundary operator** D_p as the adjoint of the boundary operator through the pairing defined previously, i.e. by definition

$$\langle\langle D_p c^{p-1}, c_p \rangle\rangle = \langle\langle c^{p-1}, A_p c_p \rangle\rangle. \quad (6)$$

Hence, $D_p = A_p^T$.

Without using components, the coboundary is defined so as to satisfy the (discrete) **Stokes Theorem**, i.e.,

$$\langle\langle dc^{p-1}, c_p \rangle\rangle = \langle\langle c^{p-1}, \partial c_p \rangle\rangle. \quad (7)$$

The **coboundary operator** (not in components) will be denoted with d .

The coboundary operator can be seen as the discrete counterpart of the exterior derivative, since if we apply the coboundary operator to a p -cochain we obtain a $(p+1)$ -cochain.

1.3 Primal and Dual Complexes

To illustrate the necessity of a dual complex in the DEC framework, we consider an example adapted from [4]. In the differential formulation of physical laws, operators such as the Laplacian act on functions to express relationships between physical quantities. For instance, the Poisson's equation

$$\Delta u(\mathbf{x}) + f(\mathbf{x}) = 0 \quad (8)$$

relates the source f at $\mathbf{x}$ to the Laplacian of u at the same point. While this might appear to be a purely pointwise relation, it is not truly local: computing the Laplacian at $\mathbf{x}$ involves evaluating derivatives, which depend on a neighborhood of $\mathbf{x}$. In other words, a differential operator inherently requires information from an extended region around each point, whose precise size is not specified. These observations naturally motivate the introduction of an auxiliary structure surrounding the original complex. Collectively, the auxiliary regions associated with each cell form a new complex, known as the **dual complex**, which provides a discrete representation of these neighborhood interactions in the framework of DEC.

Given a cell complex of dimension n , we can define its *dual* by associating a p -cell with an $(n-p)$ -cell, following a specific rule. In the case of simplicial complex, the following rules are of relevant importance, since they are the most used in applications.

- **Circumcentric (Voronoi) dual.** Voronoi cells are polygons whose sides are the axes of the edges of a primal simplicial complex.
- **Barycentric dual.** Barycentric cells are polygons obtained by connecting the barycentre of every triangle with the midpoint of the edges of the triangles.

For more details on this construction, see section 4.2 of [4]. From now on, the circumcentric rule is used to construct the dual complex.

The dual of a cell complex K is indicated as $\star K$ and a generic dual p -cell of $\star K$ is indicated as $\star\sigma_{(n-p)}$.

1.3.1 Orientations

Once a primal complex is defined, we have to orient it in order to compute properly the boundary matrices. To do so, we choose one of the two possible orientations, clockwise or counterclockwise, for all the n -cells and this choice induces an orientation for all the p -cells with $0 < p \leq n-1$, see, e.g., Definition 2.3.5. of [2]. Following [2], we do not assign any orientation to points (0-cells), even though in [4] points are oriented as sinks or sources.

Following this receipt, we can give an orientation to every p -cell of the primal complex. This induces an orientation to all the dual p -cells, since the primal p -cells and dual $(n - p)$ -cells have a clear association with each other. Precisely, following the algorithm described in Remark 2.5.1 of [2] we are able to orient all the dual p -cells.

We follow the previous rule to orient primal and dual complex. Note that this choice is crucial: if we change the orientation rule, we will also have to modify the definition of the DEC operators (coboundary, Hodge star, etc). For example, in [8] the induced orientation of the dual complex is different and consequently the definitions of some operators are slightly different.

1.3.2 Dual boundary and coboundary

We wish to define the dual boundary ∂^* in order to also have a dual coboundary d^* obtained by the relation

$$\langle \langle d^* \alpha, \star \beta \rangle \rangle = \langle \langle \alpha, \partial^* \beta \rangle \rangle. \quad (9)$$

We have to take care about the sign of the dual boundary, as stated in [9]. The following definition [9] is consistent with our orientation rule.

Definition 10. The **dual boundary operator** $\partial_p^* : \mathcal{C}_p(\star K, \mathbb{R}) \rightarrow \mathcal{C}_{p-1}(\star K, \mathbb{R})$ is the linear operator such that

$$\partial_p^* \star \tau = (-1)^{p+1} \sum_{\eta \succ \tau} \star \eta, \quad (10)$$

where $\star \tau$ is an arbitrary dual p -simplex and η is a dual $(p + 1)$ -simplex oriented in order to induce a consistent orientation on τ .

As we stated previously, the definition of the dual coboundary operator is straightforward now. The following result [10, 11], called *Poincarè duality*, gives us an explicit relation between primal coboundary and dual boundary.

Theorem 2. There exists a family of isomorphisms $\phi_p : \mathcal{C}^p(K, \mathbb{R}) \rightarrow \mathcal{C}_{n-p}(\star K, \mathbb{R})$ such that the following diagram commutes for any k .

$$\begin{array}{ccc} \mathcal{C}^{p-1}(K, \mathbb{R}) & \xrightarrow{\phi_{p-1}} & \mathcal{C}_{n-p+1}(\star K, \mathbb{R}) \\ \downarrow d_{p-1} & & \downarrow \partial_{n-p+1}^* \\ \mathcal{C}^p(K, \mathbb{R}) & \xrightarrow{\phi_p} & \mathcal{C}_{n-p}(\star K, \mathbb{R}) \end{array}$$

The proof of the previous Theorem (see proof of Theorem 65.1 of [10]) is constructive and we can see that the definition of ϕ_p is dependent on the orientation.

The following result is a practical consequence of the previous statements [9, 3].

Theorem 3. Given a primal n -complex, the matrix representation of the boundary operator on primal p -cells A_p is the transpose of the (matrix representation of the) boundary operator on dual $(n - p - 1)$ -cells A_{n-p-1}^* , i.e.

$$A_p = (-1)^{p+1} (A_{n-p-1}^*)^T, \quad (11)$$

or equivalently

$$A_{n-p-1}^* = (-1)^{p+1} A_p^T \quad (12)$$

All these relations can be summarized by the following diagram, in which $\star$ is the **discrete Hodge star** (introduced formally in the next section) which allows us to pass from a primal p -cell to its correspondent dual $(n - p)$ -cell.

$$\begin{array}{ccc} \mathcal{C}^p(K, \mathbb{R}) & \xrightarrow{d_p} & \mathcal{C}^{p+1}(K, \mathbb{R}) \\ \downarrow \star_p & & \downarrow \star_{p+1} \\ \mathcal{C}^{n-p}(\star K, \mathbb{R}) & \xleftarrow{d_{n-p-1}^* = (-1)^{p+1} d_p^t} & \mathcal{C}^{n-p-1}(\star K, \mathbb{R}) \end{array}$$

Example 4. To get in touch with the relation between primal and dual coboundary, consider the following example (Figure 3).

We first orient every primal 2-cell counterclockwise, while every primal 1-cell inherits the orientation of the parent 2-cell (according to Definition 2.3.5 in [2]), except σ_4^1 , in which we choose an arbitrary orientation. Now we are able to orient the dual 1 and 2-cells consistently with respect to primal orientations. We follow the algorithm described in Remark 2.5.1 in [2]. To see how it works, we orient for example the dual 2-cell $\star \sigma_0^0$ and the dual 1-cell $\star \sigma_1^1$.

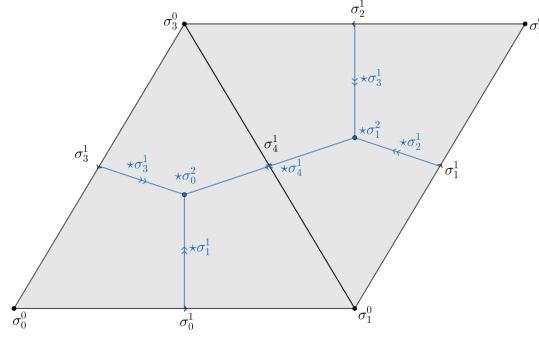


Figure 3: Primal cell complex (grey) and its dual (blue). Primal 2-cells are oriented counterclockwise, inducing an orientation on the 1-cells that are not shared by any 2-cell. The dual complex is oriented following the algorithm described in Remark 2.5.1 in [2]. We remark that the orientation of each oriented dual edge can be obtained by rotating of $\pi/2$ counterclockwise the corresponding primal edge [7].

1. We have to understand if $\star\sigma_0^0$ is oriented counterclockwise or clockwise. To do so, following [2] and his notation we have to compute

$$s := \text{sgn}([c(\sigma_0^0), c(\sigma_1^0), c(\sigma_2^0)], \sigma_2^0)$$

A simple computation shows that $s = 1$, thus the dual simplex $\star\sigma_0^0$ is oriented counterclockwise. Indeed, we have that $\star\sigma_0^0 = s[c(\sigma_0^0), c(\sigma_1^0), c(\sigma_2^0)] = [c(\sigma_0^0), c(\sigma_1^0), c(\sigma_2^0)]$.

2. For sure $\star\sigma_1^1 = s[c(\sigma_1^0), c(\sigma_2^0)]$, where $s = \pm 1$. According to Remark 2.5.1 of [2], we actually have

$$s = \text{sgn}([c(\sigma_0^0), c(\sigma_1^0)], \sigma_1^0) \cdot \text{sgn}([c(\sigma_0^0), c(\sigma_1^0), c(\sigma_2^0)], \sigma_2^0) = 1 \cdot 1 = 1.$$

Hence, $\star\sigma_1^1 = [c(\sigma_1^0), c(\sigma_2^0)]$.

This algorithm gives us an orientation of the dual complex. In this way, we can compute the boundary primal matrix A_1 and the boundary dual matrix $A_2^\star$. Following the definition of boundary, we get

$$A_1 = \begin{bmatrix} -1 & 0 & 0 & 1 & 0 \\ 1 & -1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 1 \end{bmatrix}, \quad \star A_2 = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{bmatrix}.$$

Hence, we have $A_2^\star = -A_1^T$.

1.4 The Discrete Hodge Star

The continuous Hodge star is an isomorphism between p -forms and $(n-p)$ -forms. Similarly, in the discrete setting it is an isomorphism between primal p -cochains and dual $(n-p)$ -cochains.

Definition 11. The **discrete Hodge star** is the map $\star : \mathcal{C}^p(K, \mathbb{R}) \rightarrow \mathcal{C}^{n-p}(\star K, \mathbb{R})$ defined by the following relation: for any p -cochain c^p and p -simplex σ_p

$$\frac{1}{|\sigma_p|} \langle c^p, \sigma_p \rangle = \frac{1}{|\star\sigma_p|} \langle \star c^p, \star\sigma_p \rangle, \quad (13)$$

where $\star\sigma_p$ is the (only) dual $(n-p)$ -simplex corresponding to σ_p , and $|\sigma_p|$ (respectively $|\star\sigma_p|$) is the volume of the primal p -simplex (respectively dual $(n-p)$ -simplex).

As in the continuous theory, the following relation holds [6]:

$$\star_p \star_{n-p} = (-1)^{p(n-p)} \text{Id}, \quad (14)$$

where, with a slight abuse of notation, we use the symbol $\star$ both for the Hodge star and its inverse. In Figure 4 (inspired from the first figure of Section 4.8.4 in [7]) we show graphically the action of this map in some simple cases.

We can now easily translate the definition of the L^2 inner product in the discrete setting.

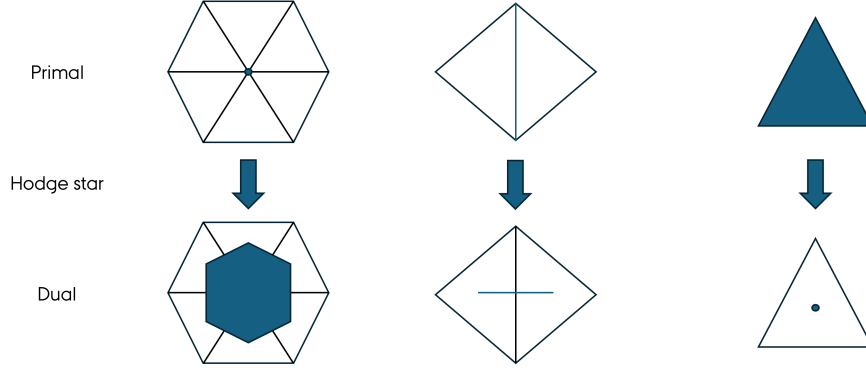


Figure 4: The action of the Hodge star.

Definition 12. Let $c_1^p, c_2^p \in \mathcal{C}^p(K, \mathbb{R})$, their **discrete L^2 inner product** is defined as

$$\langle c_1^p, c_2^p \rangle := \sum_{\sigma^p \in K} \langle \langle c_1^p, \sigma^p \rangle \rangle \langle \langle \star c_2^p, \star \sigma^p \rangle \rangle \quad (15)$$

In matrix-vector form the previous definition becomes

$$c_1^T S c_2. \quad (16)$$

In the same fashion, it is possible to define the discrete L^2 inner product for dual p -cochains. In [3] the discrete L^2 inner product of $c_1^p, c_2^p \in \mathcal{C}^p(K, \mathbb{R})$ is defined as

$$\langle c_1^p, c_2^p \rangle := \sum_{\sigma^p \in K} \frac{|V_{\sigma^p}|}{|\sigma^p| |\star \sigma^p|} \langle \langle c_1^p, \sigma^p \rangle \rangle \langle \langle \star c_2^p, \star \sigma^p \rangle \rangle = \frac{1}{n} \sum_{\sigma^p \in K} \langle \langle c_1^p, \sigma^p \rangle \rangle \langle \langle \star c_2^p, \star \sigma^p \rangle \rangle, \quad (17)$$

which becomes in matrix-vector form

$$\frac{1}{n} c_1^T S c_2. \quad (18)$$

We prefer to define the discrete inner product as in (15). Indeed, “mimicking the smooth setting, each inner product matrix S determines a linear mapping from a discrete p -form to a discrete version of the dual $(n-p)$ -form” (hence a discrete Hodge star operator!) [12]. In other words, the second definition is not consistent with our definition of the Hodge Star since, for $n \neq 1$, $\frac{1}{n} S \neq S$.

1.5 The Discrete Laplace-de Rham Operator

To define the discrete version of the Laplace-de Rham Operator we need the concept of discrete codifferential, following the derivation obtained in the continuous setting.

Definition 13. The **discrete codifferential operator** δ_p maps *primal* p -cochains into *primal* $(p-1)$ -cochains via a mapping into the *dual* complex and is defined as

$$\delta_p := (-1)^{n(p-1)+1} \star d_{n-p}^* \star. \quad (19)$$

Moreover, by definition $\delta_0 \equiv 0$.

In the previous definition, we write $\star$ both for primal to dual map and dual to primal map. The definition above is obtained requiring that, for any $\alpha \in \mathcal{C}^{p-1}(K, \mathbb{R}), \beta \in \mathcal{C}^p(K, \mathbb{R})$

$$\langle d\alpha, \beta \rangle = \langle \alpha, \delta\beta \rangle. \quad (20)$$

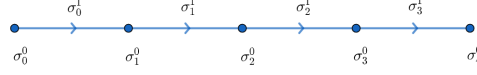


Figure 5: An example for $m = 4$ of the 1-simplicial complex considered in Example 5.

We can define similarly the discrete dual codifferential operator as a map from dual p -cochains to dual $(p - 1)$ -cochains as follows. We require that for any $\alpha \in \mathcal{C}^{p-1}(\star K, \mathbb{R})$, $\beta \in \mathcal{C}^p(\star K, \mathbb{R})$

$$\langle d^* \alpha, \beta \rangle = \langle \alpha, \delta^* \beta \rangle. \quad (21)$$

Performing all the computations and exploiting the previous equation we finally get

$$\delta_p^* := (-1)^{p(n+1-p)} \star d_{n-p} \star. \quad (22)$$

Definition 14. The **discrete Laplace-de Rham Operator** is defined as

$$\Delta := d\delta + \delta d. \quad (23)$$

1.6 de Rham Maps and Discrete Flat Operators

The de Rham maps establish a connection between smooth differential forms and cochains. These maps encode how continuous geometric and physical quantities, such as fields or fluxes, can be integrated over simplices in a simplicial complex.

In the following, let M be a smooth triangulable n -manifold and K be a simplicial complex homeomorphic to M , as in [2]. In the following, $\Lambda^p(M)$ is the space of continuous p -form on M .

Definition 15. The **de Rham map** is the map $\mathcal{R}^p : \Lambda^p(M) \rightarrow \mathcal{C}^p(K, \mathbb{R})$ defined as

$$\langle \langle \mathcal{R}^p(\omega), c_p \rangle \rangle := \sum_i a_i \int_{\sigma_i^p} \omega, \quad (24)$$

where $c_p = \sum_i a_i \sigma_i^p$ is a generic p -chain. The definition of the **dual de Rham map** is analogous.

This definition associates to each smooth differential form a discrete cochain by integrating the form over simplices. In this way, continuous geometric data are transferred into an algebraic setting where they act linearly on chains, providing a natural link between differential and discrete forms.

To connect discrete data with continuous geometric structures, interpolation operators are used to reconstruct smooth fields from discrete differential forms.

Definition 16. A **primal interpolation** is a function $\mathcal{I}^1 : \mathcal{C}^0(K, \mathbb{R}) \rightarrow \Lambda^1(M)$ and similarly a **dual interpolation** is a function $\mathcal{I}^{*1} : \mathcal{C}^0(\star K, \mathbb{R}) \rightarrow \Lambda^1(M)$.

Definition 17. Given a primal interpolation function $\mathcal{I}^1$, we define the **discrete primal 1-flat** b_1 as the function $b_1 : \mathcal{C}^0(K, \mathbb{R}) \rightarrow \mathcal{C}^1(K, \mathbb{R})$ such that

$$b_1(c^0) := \mathcal{R}^1(\mathcal{I}^1(c^0)). \quad (25)$$

The definition of the **discrete dual 1-flat** is analogous.

The idea of this definition comes from [2] and [13], the difference being that we explicitly separate the contribution of interpolation from the process of integration. We remark that in this definition we are identifying primal/dual 0-cochains and discrete (one-dimensional) vector fields¹, so that the discrete 1-flat can be identified as a map between discrete fields and forms (as in the continuous framework).

Example 5 (Some flat operators in 1D). Consider a 1-simplicial complex K with $m + 1$ vertices and m edges. An example with $m = 4$ is provided in Figure 5. Let $\{\sigma_0^0, \dots, \sigma_m^0\}$ and $\{\sigma_0^1, \dots, \sigma_{m-1}^1\}$ be respectively the sets of vertices and edges of K . Note that in this case, the manifold M can be defined as an open (continuous) interval containing the 1-simplicial complex K . To keep the notation concise, in this context we will use the same notation for the 0-simplices and its (1D) coordinates.

¹ For a formal definition of a discrete vector field, see Definition 5.1.1-5.1.2 in [2].

² To be precise, this is not rigorous: we are going to define interpolation function that return forms which are differentiable in the interior of each 1-simplex.

PARABOLIC FLAT In this case, we define, for $x \in (\sigma_i^0, \sigma_{i+1}^0)$,

$$\mathcal{I}^1(c^0) := \left(\frac{c^0(\sigma_0^{i+1}) - c^0(\sigma_0^i)}{\sigma_{i+1}^0 - \sigma_i^0} x + c^0(\sigma_0^i) - \frac{c^0(\sigma_0^{i+1}) - c^0(\sigma_0^i)}{\sigma_{i+1}^0 - \sigma_i^0} \right) dx, \quad (26)$$

using the notation introduced in (4). Thus, we have

$$b_{par}^1(c^0)(\sigma_1^i) = \int_{\sigma_i^1} \mathcal{I}^1(c^0) \quad (27)$$

$$= \int_{\sigma_i^0}^{\sigma_{i+1}^0} \left(\frac{c^0(\sigma_0^{i+1}) - c^0(\sigma_0^i)}{\sigma_{i+1}^0 - \sigma_i^0} x + c^0(\sigma_0^i) - \frac{c^0(\sigma_0^{i+1}) - c^0(\sigma_0^i)}{\sigma_{i+1}^0 - \sigma_i^0} \right) dx \quad (28)$$

$$= \frac{c^0(\sigma_0^i) + c^0(\sigma_0^{i+1})}{2} (\sigma_{i+1}^0 - \sigma_i^0). \quad (29)$$

DOWNWIND FLAT We define for $x \in (\sigma_i^0, \sigma_{i+1}^0)$,

$$\mathcal{I}^1(c^0) := c^0(\sigma_0^{i+1}) dx, \quad (30)$$

which gives

$$b_{dow}^1(c^0)(\sigma_1^i) = \int_{\sigma_i^1} \mathcal{I}^1(c^0) = c^0(\sigma_0^{i+1})(\sigma_{i+1}^0 - \sigma_i^0). \quad (31)$$

UPWIND FLAT Similarly, we define for $x \in (\sigma_i^0, \sigma_{i+1}^0)$,

$$\mathcal{I}^1(c^0) := c^0(\sigma_0^i) dx, \quad (32)$$

which gives

$$b_{upw}^1(c^0)(\sigma_1^i) = \int_{\sigma_i^1} \mathcal{I}^1(c^0) = c^0(\sigma_0^i)(\sigma_{i+1}^0 - \sigma_i^0). \quad (33)$$

1.7 Vector-Valued Cochains

Until now, greater attention was given to the discrete counterpart of scalar differential forms. However, in the continuous setting we frequently face cases where we have to manipulate vector and tensor fields. Thus, we need to translate this framework properly in the discrete scenario. In other words, a generalization of the definition of cochain is needed.

Definition 18. A **vector-valued p -cochain** is the group of linear functions from $\mathcal{C}_p(K, \mathbb{R})$ to a vector space V and it is indicated as $\mathcal{C}^p(K, V)$. In the same fashion, let Lin be the vector space of tensors. A **tensor-valued p -cochain** is the group of linear functions from $\mathcal{C}_p(K, \mathbb{R})$ to Lin and it is indicated as $\mathcal{C}^p(K, \text{Lin})$.

Any vector-valued p -cochain can be written as a linear combination (with vector coefficients!) of basic p -cochains, i.e., we can write a generic vector-valued p -cochain as

$$\mathbf{c}^p = \sum_i \mathbf{c}^p(\sigma_p^i) \sigma_p^i, \quad (34)$$

where each $\mathbf{c}^p(\sigma_p^i) \in \mathcal{V}$.

Reasoning in components, let $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ be a base of a vector space V . A vector valued p -cochain $\mathbf{c}^p$ maps a p -chain c_p into a vector $\mathbf{c}^p(c_p) = \mathbf{c}_i^p \mathbf{e}_i$. Note that $\mathbf{c}_i^p$ is a linear map from p -chains to scalars, i.e. a scalar-valued p -cochain! An analogous analysis can be done for tensor-valued cochains.

Notice that all the discrete theory developed so far should be properly extended in this case. Some operations are straightforward to extend (e.g., the inner product), but for others an extended differential theory is needed. For example, the coboundary of vector valued-cochains can be seen as the discrete version of the exterior covariant derivative [14]. Since not all the operators will be of our interest for vector-valued cochains, we are not going to present their formal extension here. For further information, we refer the reader to [14].

2 APPLICATION OF DEC TO COMPUTATIONAL PHYSICS: EXAMPLES

This section presents a series of examples demonstrating the application of DEC to classical problems in computational physics. We showcase how DEC provides a powerful and flexible framework for building the governing equations of several physical systems on discrete manifolds.

2.1 2D Poisson Equation

Here, we derive the stationary diffusion equation with source term (*i.e.*, the *Poisson Equation*) using the tools of discrete calculus seen in the previous sections. We remark that our formulation is inherently discrete, *i.e.* we *do not* discretize the continuous Poisson equation.

In 2D, let us consider a primal 2-complex consisting of n vertices. We assign an orientation of the primal complex and the dual complex as described in Section 1.3. In particular, we orient every primal 2-cell counterclockwise and we find, by applying the algorithm of Remark 2.5.1 in [2], that also dual 2-cells are oriented counterclockwise. A portion of an example of the 2D simplicial complex is represented in Figure 6.

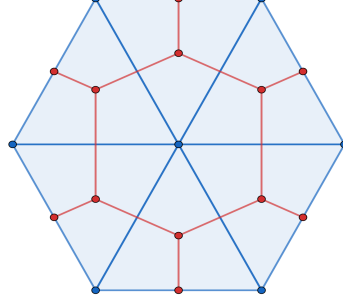


Figure 6: A portion of the 2D simplicial cell complex considered (light blue) and its *circumcentric* dual (red).

We represent the diffusing quantity as a primal (scalar-valued) 0-cochain denoted by the symbol u . Applying the coboundary operator d_0 we get a primal scalar-valued 1-cochain d_0u in which the scalars attached to every basic 1-cochain are the discrete equivalent of the gradient of the continuous scalar field integrated along the edge. To pass from a circulation to a flux across the edges, we apply the Hodge star operator $\star_1$, which provides a scalar-valued dual 1-cochain $\star_1 d_0u$. Then, using a linear and isotropic constitutive equation to relate the diffusive flux h to $\star_1 d_0u$, we write

$$h = -k \star_1 d_0u, \quad (35)$$

where k is a (constant) material parameter (conductivity/diffusivity). Since the dual 2-cells are oriented counterclockwise, following [4] the positive orientation for h describes the flux *leaving* any dual 2-cell.

To obtain the *net* flux *entering* any dual 2-cell we have to add up the contributions along its boundary using the dual coboundary operator d^* ($= -d_0^t$, see Section 2.3) and then change the sign of the result. If f is a known dual 2-cochain which describes the external sources of the dual 2-cells, the balance equation reads

$$-d^*h + f = 0. \quad (36)$$

Using the constitutive equation for the flux

$$k d_0^t(\star_1 d_0u) + f = 0. \quad (37)$$

To recover the formulation of diffusion equation with discrete Laplacian, recall that

$$\Delta := \delta d + d\delta. \quad (38)$$

Since u is a 0-cochain, $d\delta u = 0$ by definition of δ . Moreover, since $\delta_1 = -\star d^*\star$, applying the (inverse of) Hodge star operator (we again write $\star$ both for primal to dual map and dual to primal map) to (37) we get

$$-k\Delta u + \tilde{f} = 0, \quad (39)$$

where $\tilde{f} := \star f$. Since the Laplacian acting on fields ∇^2 is minus the Laplacian on forms (see [15]), the last equation is the discrete analogous of the Poisson equation $k\nabla^2 u + \tilde{f} = 0$ (where u and $\tilde{f}$ are scalar fields).

In the continuous setting, one can show [16] that the Poisson equation arises as the stationarity conditions of the Dirichlet functional, which comprises two terms: the first is the L^2 norm of the gradient of the unknown field, and the second is the L^2 inner product between the unknown field and the source term. Similarly, in the discrete setting, a discrete Dirichlet energy can be defined by representing the dimensionless unknown u and the source f as primal scalar-valued 0-cochains:

$$\mathcal{E}_p(u) := \frac{1}{2} \langle du, du \rangle - \langle u, f \rangle = \frac{1}{2} \langle \delta du, u \rangle - \langle u, f \rangle. \quad (40)$$

Here, du is the coboundary of u , a primal scalar-valued 1-cochain whose entries correspond to the integral of the continuous gradient field along the mesh edges. Consequently, each term in (40) preserves the same physical and geometric interpretation as in the continuous formulation.

2.2 Euler's Elastica

In the following, $\mathcal{I} = [0, L] \subset \mathbb{R}$ is a one-dimensional interval that represents the coordinates of the points of the longitudinal axis of the rod in its undeformed, straight configuration, L is the length of the rod, $u(\sigma)$ is the angle formed by the tangent to the axis of the rod and the horizontal direction at the dimensionless arc-length $\sigma := s/L$. For a constant cross-section rod subject to a vertical end-load at the right end and clamped at the left end, the boundary-value problem for the (dimensionless) continuous Euler's Elastica equation reads [17]

$$\frac{d^2 u}{d\sigma^2} + f \cos u = 0 \quad \text{in } [0, 1], \quad u(0) = 0, \quad \frac{du}{d\sigma}(1) = 0. \quad (41)$$

In (41), $f = PL^2/B$ is the dimensionless load parameter, where $P < 0$ is the vertical component of the load applied at the right end and B is the *bending stiffness* of the rod. The variational formulation of (41) is

$$\min_u \frac{1}{2} \int_0^1 \left(\frac{du}{d\sigma} \right)^2 d\sigma - \int_0^1 f \sin u d\sigma, \quad u(0) = 0. \quad (42)$$

To derive the discrete version of (41) and (42), consider a 1-simplicial complex representing a discrete rod. The discrete elastica energy can be formulated as

$$\mathcal{E}_{\text{el}}(u) := \frac{1}{2} \langle k, \star d^\star u \rangle - \langle f \mathbb{1}, \sin u \rangle, \quad (43)$$

where $\mathbb{1}$ is the dual one 0-cochain, $k := \mathbb{1}_{\text{int}} \odot \star d^\star u$ is the primal 0-cochain that encodes the (discrete) curvatures on the nodes and $\mathbb{1}_{\text{int}}$ is the primal 0-cochain that is identically 1 on the interior and 0 on the boundary nodes. We can re-write (43) as

$$\mathcal{E}_{\text{el}}(u) = \frac{1}{2} \langle \star(\mathbb{1}_{\text{int}} \odot \star d^\star u), d^\star u \rangle - \langle f \mathbb{1}, \sin u \rangle \quad (44)$$

$$= \frac{1}{2} \langle \delta^\star \star(\mathbb{1}_{\text{int}} \odot \star d^\star u), u \rangle - \langle f \mathbb{1}, \sin u \rangle \quad (45)$$

$$= \frac{1}{2} \langle \delta^\star \star k, u \rangle - \langle f \mathbb{1}, \sin u \rangle, \quad (46)$$

Computing the first variation of the discrete functional we get

$$\delta^\star \star k - f \cos(u) = 0, \quad (47)$$

which is equivalent, using the definition of the codifferential, to

$$- \star dk - f \cos(u) = 0. \quad (48)$$

2.3 2D Linear Elasticity

Let K be a 2-simplicial complex, $\hat{\mathbf{e}}_i$ be the Cartesian coordinate framework, $\mathbf{e}_i$ be the reference local basis of a given 2-simplex and $\mathbf{e}'_i$ be the local basis of the deformed 2-simplex. The contravariant bases are indicated with upper indexing. Denoting with $\mathbf{F}$ the deformation gradient of a given 2-simplex we have³

$$\mathbf{F} = \mathbf{e}'_i \otimes \mathbf{e}^i. \quad (49)$$

Indeed, for any k we have

$$\mathbf{F} \mathbf{e}_k = (\mathbf{e}'_i \otimes \mathbf{e}^i) \mathbf{e}_k = \mathbf{e}'_i (\mathbf{e}^i \cdot \mathbf{e}_k) = \mathbf{e}'_k. \quad (50)$$

Now recall that

$$\mathbf{e}'_i = (\mathbf{e}'_i)_j \hat{\mathbf{e}}_j, \quad \mathbf{e}_k = (\mathbf{e}'_k)_l \hat{\mathbf{e}}_l, \quad \mathbf{e}^i = \mathbf{g}_R^{ik} \mathbf{e}_k, \quad (51)$$

where $\mathbf{g}_R^{ik} := \mathbf{e}^i \cdot \mathbf{e}^k$. Putting everything together in (49) we get

$$\mathbf{F} = (\mathbf{e}'_i)_j \hat{\mathbf{e}}_j \otimes \mathbf{g}_R^{ik} (\mathbf{e}_k)_l \hat{\mathbf{e}}_l = (\mathbf{e}'_i)_j \mathbf{g}_R^{ik} (\mathbf{e}_k)_l \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_l. \quad (52)$$

³ We adopt Einstein notation for brevity.

In particular, in matrix notation we have

$$\mathbf{F} = (\mathbf{A}')^T \mathbf{g}_R \mathbf{A}, \quad (53)$$

where $\mathbf{A}$ (resp. $\mathbf{A}'$) is the matrix where the i -th rows is the i -th basis vector $\mathbf{e}_i$ (resp. $\mathbf{e}'_i$). Thus, we may represent the deformation gradient $\mathbf{F}$ as a dual (tensor-valued) zero-cochain. Now, we define the 2D infinitesimal strain measure as the following dual zero-cochain

$$\mathbf{E} := \frac{1}{2}(\mathbf{F} + \mathbf{F}^T) - \mathbf{I}, \quad (54)$$

where $\mathbf{I}$ is the identity dual zero-cochain. By applying the constitutive equation for linearly elastic isotropic bodies, we can compute the dimensionless stress as a dual zero-cochain

$$\mathbf{S} := 2\mathbf{E} + \frac{\lambda}{\mu} \text{tr}(\mathbf{E})\mathbf{I}, \quad (55)$$

where (μ, λ) are the Lamé moduli. Note that when computing the stress cochain $\mathbf{S}$ we are multiplying (coefficient-wise) a scalar-valued ($\text{tr}(\mathbf{E})$) and a tensor-valued ($\mathbf{I}$) cochain.

In the case of zero body forces and Dirichlet boundary conditions, we easily derive the potential energy as

$$\mathcal{E}_{\text{LE}} := \frac{1}{2} \langle \mathbf{S}, \mathbf{E} \rangle. \quad (56)$$

In case of non-zero body forces, (56) becomes

$$\mathcal{E}_{\text{LE}} := \frac{1}{2} \langle \mathbf{S}, \mathbf{E} \rangle - \langle \mathbf{u}, \mathbf{f} \rangle, \quad (57)$$

where $\mathbf{u}$ and $\mathbf{f}$ are the primal vector-valued zero cochains representing, respectively, the dimensionless displacements of the primal nodes and the body force at each primal node.

2.4 Burgers' equation

Finite Volume (FV) schemes and DEC share a deep structural connection, as both rely on integral formulations of conservation laws over discrete domains. In FV methods, fluxes are computed across control volume interfaces to ensure local conservation. DEC, on the other hand, provides a coordinate-free framework for discretizing differential forms, naturally encoding fluxes through discrete Stokes' theorem. This correspondence reveals that many FV schemes can be interpreted as specific realizations of DEC, highlighting their common geometric and conservation principles. In this section, we show this link through an illustrative example, the Burger's equation, which is closely related to the first-order traffic flow models.

In the continuous setting, the general form of the one-dimensional Burgers' equation, also called *viscous Burgers' equation*, reads

$$u_t = -uu_x + \nu u_{xx}, \quad (58)$$

where $u(x, t)$ is the speed of fluid in (x, t) and ν is the viscosity of the fluid. The *inviscid Burgers' equation* is obtained by setting $\nu = 0$ and it is a prototype for conservation equations that can develop *shocks*

$$u_t = -(u^2/2)_x. \quad (59)$$

In the discrete setting, a 1-simplicial complex models the space, while the time interval is uniformly discretized with step-size Δ_t . The unknown U^n is the average value of the velocity field at the time $t = n\Delta_t$ in a given dual edge and it is represented as a dual 0 cochain. The flux F is a primal 0-cochain such that

$$F_{i-1/2}^n := f(U_{i-1}^n, U_i^n),$$

where f is some numerical flux function (as in the Finite Volume Method schemes). Forward Euler method is chosen to treat the term u_t . The discrete version of the Burgers' equation reads in general

$$U_i^{n+1} = U_i^n - \Delta_t (\star dF^n)_i. \quad (60)$$

The choice of the numerical flux function leads to different numerical schemes. For the viscid Burgers' equation, a popular choice is the *parabolic scheme*, i.e.

$$F^n := \star b_{\text{par}}^1(-(U^n)^2/2) + \nu \star dU^n, \quad (61)$$

where $b_{\text{par}}^1 = \mathcal{R}^{*0}(I_{\text{par}}^1)$ is the one defined in Example 5. Explicitly,

$$f(U_{i-1}^n, U_i^n) = -\frac{(U_{i-1}^n)^2 + (U_i^n)^2}{4} + \nu \frac{U_i^n - U_{i-1}^n}{|\star \sigma_{0,i}|}. \quad (62)$$

The parabolic scheme in DEC formulation is equivalent to the Finite Difference Method where the nodes are the dual 0-simplices of the complex. Instead, for the inviscid Burgers' equation we use the *upwind scheme* (more stable than the parabolic scheme), i.e.

$$F^n := \star b_{\text{upw}}^1(-(U^n)^2/2), \quad (63)$$

where $b_{\text{upw}}^1 = \mathcal{R}^{*0}(I_{\text{up}}^1)$ and I_{upw} is the constant interpolation defined in Example 5. Explicitly,

$$f(U_{i-1}^n, U_i^n) = -(U_{i-1}^n)^2/2. \quad (64)$$

Finally, notice that the dimensionless Burgers' equation is obtained defining the (dimensionless) variables

$$\hat{x} := \frac{x}{L}, \quad \hat{t} := \frac{t}{T}, \quad \hat{u} := \frac{u}{U}, \quad (65)$$

where L is a characteristic length, U is a characteristic velocity and $T := L/U$. It reads

$$\hat{u}_{\hat{t}} = -\hat{u}\hat{u}_{\hat{x}} + \hat{\nu}\hat{u}_{\hat{x}\hat{x}}, \quad (66)$$

where $\hat{\nu} := \nu/LU$.

REFERENCES

- [1] R. Abraham, J. E. Marsden, and T. Ratiu, *Manifolds, Tensor Analysis, and Applications*, vol. 75. Springer Science & Business Media, 2012.
- [2] A. N. Hirani, *Discrete Exterior Calculus*. California Institute of Technology, 2003.
- [3] M. Desbrun, A. N. Hirani, M. Leok, and J. E. Marsden, "Discrete Exterior Calculus," *arXiv preprint math/0508341*, 2005.
- [4] E. Tonti, *The Mathematical Structure of Classical and Relativistic Physics*, vol. 10. Springer, 2013.
- [5] M. Griebel, C. Rieger, and A. Schier, "Upwind Schemes for Scalar Advection-Dominated Problems in the Discrete Exterior Calculus," *Transport Processes at Fluidic Interfaces*, pp. 145–175, 2017.
- [6] E. Grinspun, M. Desbrun, K. Polthier, P. Schröder, and A. Stern, "Discrete Differential Geometry: An Applied Introduction," *ACM Siggraph Course*, vol. 7, no. 1, 2006.
- [7] K. Crane, "Discrete Differential Geometry: An Applied Introduction," *Notices of the AMS, Communication*, pp. 1153–1159, 2018.
- [8] L. J. Grady and J. R. Polimeni, *Discrete Calculus: Applied analysis on graphs for computational science*, vol. 3. Springer, 2010.
- [9] E. Schulz and G. Tsogtgerel, "Convergence of Discrete Exterior Calculus Approximations for Poisson Problems," *Discrete & Computational Geometry*, vol. 63, pp. 346–376, 2020.
- [10] J. R. Munkres, *Elements of Algebraic Topology*. CRC press, 2018.
- [11] F. Milicchio, A. DiCarlo, A. Paoluzzi, and V. Shapiro, "A codimension-zero approach to discretizing and solving field problems," *Advanced Engineering Informatics*, vol. 22, no. 2, pp. 172–185, 2008.
- [12] F. De Goes, M. Desbrun, M. Meyer, and T. DeRose, "Subdivision Exterior Calculus for Geometry Processing," *ACM Transactions on Graphics (TOG)*, vol. 35, no. 4, pp. 1–11, 2016.
- [13] P. D. Boom, O. Kosmas, L. Margetts, and A. P. Jivkov, "A geometric formulation of linear elasticity based on discrete exterior calculus," *International Journal of Solids and Structures*, vol. 236, p. 111345, 2022.

- [14] D. Berwick-Evans, A. N. Hirani, and M. D. Schubel, "Discrete Vector Bundles with Connection and the Bianchi Identity," *arXiv preprint arXiv:2104.10277*, 2021.
- [15] T. Frankel, *The Geometry of Physics: An Introduction*. Cambridge University Press, 2011.
- [16] L. C. Evans, *Partial Differential Equations*, vol. 19. American Mathematical Society, 2010.
- [17] B. Audoly and Y. Pomeau, *Elasticity and Geometry*. Oxford University Press, 2018.